Paleomagnetic Study of Recent Cascade Lavas, Three Sisters, Oregon

DONALD F. HEINRICHS[1]

School of Oceanography, Oregon State University, Corvallis, Oregon 97331

The natural remanent magnetization of 94 oriented cores from nine recent lava flows was measured. The field area is located in the Three Sisters area of the Oregon Cascades. The maximum possible age range for the flows is from 300 to 12,000 years B.P.; eight of the nine flows are younger than 3850 years. The mean direction of magnetization ($D = 2.2°$, $I = +49.6°$, $\alpha_{95} = 5.2°$) is statistically distinct from the axial geomagnetic dipole field ($D = 0°$, $I = +62.9°$). It is not clear whether this mean direction is a significant departure of the main geomagnetic field from an axial direction or part of the 'normal' secular change.

To a close approximation, paleomagnetic results from Recent and Quaternary rocks show the earth's geomagnetic field to have been that of a geocentric axial dipole [*Hospers*, 1955; *Irving*, 1964; *Opdyke and Henry*, 1969]. In contrast to other investigations of Recent rocks, in which the mean paleomagnetic field directions are coincident with those of an axial dipole, the mean direction found here diverges from an axial dipole direction. The magnetic inclination of each individual flow mean direction of magnetization is consistently too shallow; the declinations center around geographic north, however. After demagnetization, eight of the nine individual flow mean directions are statistically distinct from both the present geomagnetic field direction and a geocentric axial dipole field direction.

The field area is located in the Three Sisters area of the Oregon Cascades between the Pleistocene volcanoes known as Three Fingered Jack and North Sister (Figure 1). Numerous cinder cones stand in the midst of Recent basaltic lava fields whose total area exceeds 240 km^2. *Williams* [1944, 1957] published two geologic reconnaissance studies of the area. *Taylor* [1965, 1967] extensively surveyed the individual lava flows and determined their eruptive history. The geologic terminology used in this paper is based on Taylor's usage. *Taylor* [1967] defines three major lava fields in the area: Sand Mountain lava field, Belknap lava field, and the Yapoah–Four-in-One–Collier lava field. The most extensive unit is the Belknap lava field, which forms a shield volcano produced by flows from Belknap crater, Little Belknap volcano, and South Belknap cone (Figure 1). The fresh rock of the lava flows is generally a black, glassy, vesicular basalt containing scattered phenocrysts of plagioclase and olivine with a range of compositions from olivine basalt to basaltic andesite [*Taylor*, 1967]. The flows show little evidence of weathering or alteration. The solid parts of the flows lie several meters beneath an outer surface covered by jagged blocks and flow breccia and are accessible only in such places as lava tubes, lava gutters, road cuts, and quarries. Some of the flows are largely blocky with only occasional continuity of the basal parts. A total of 94 cores was drilled from the nine flows. The maximum separation between sample sites within a given flow was approximately 1 km, and the average distance between samples at each site was 7 meters.

The ages of four of the flows investigated have been found directly by radiocarbon dating (Table 1). Two dates for Clear Lake flow [*Benson*, 1965] were obtained from a submerged tree in Clear Lake. The other five dates listed, including an additional date for the Clear Lake flow, were obtained from carefully chosen charcoal samples from trees that were engulfed by the hot lavas or cinders. In addition, structural relationships between the flows, together with radiocarbon dates on other units not sampled

[1] Now at Office of Naval Research, Marine Geology and Geophysics, Arlington, Virginia 22217.

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(mostly cinders), enable the ages of the remaining five flows to be estimated [*Taylor*, 1967].

MEASUREMENTS

The directions and intensities of remanent magnetization were measured on a spinner magnetometer similar in basic design to the instrument described by *Doell and Cox* [1965]. The direction measurements are reproducible to ±1.2° (s.d.) with an estimated accuracy of ±1.5° (s.d.). An instrument modification improved the precision and accuracy of the intensity determinations for the later measurements; however, in all cases the estimated accuracy is within ±10%. Initial demagnetization experiments were made by a two-axis tumbler with an alternating magnetic field produced by a pair of Helmholtz coils. The later measurements were made by a four-axis tumbler with the current in the coils controlled by a variable transformer. At least one specimen per flow was progressively demagnetized to 800 oe to determine the optimum field required to remove secondary components of magnetization.

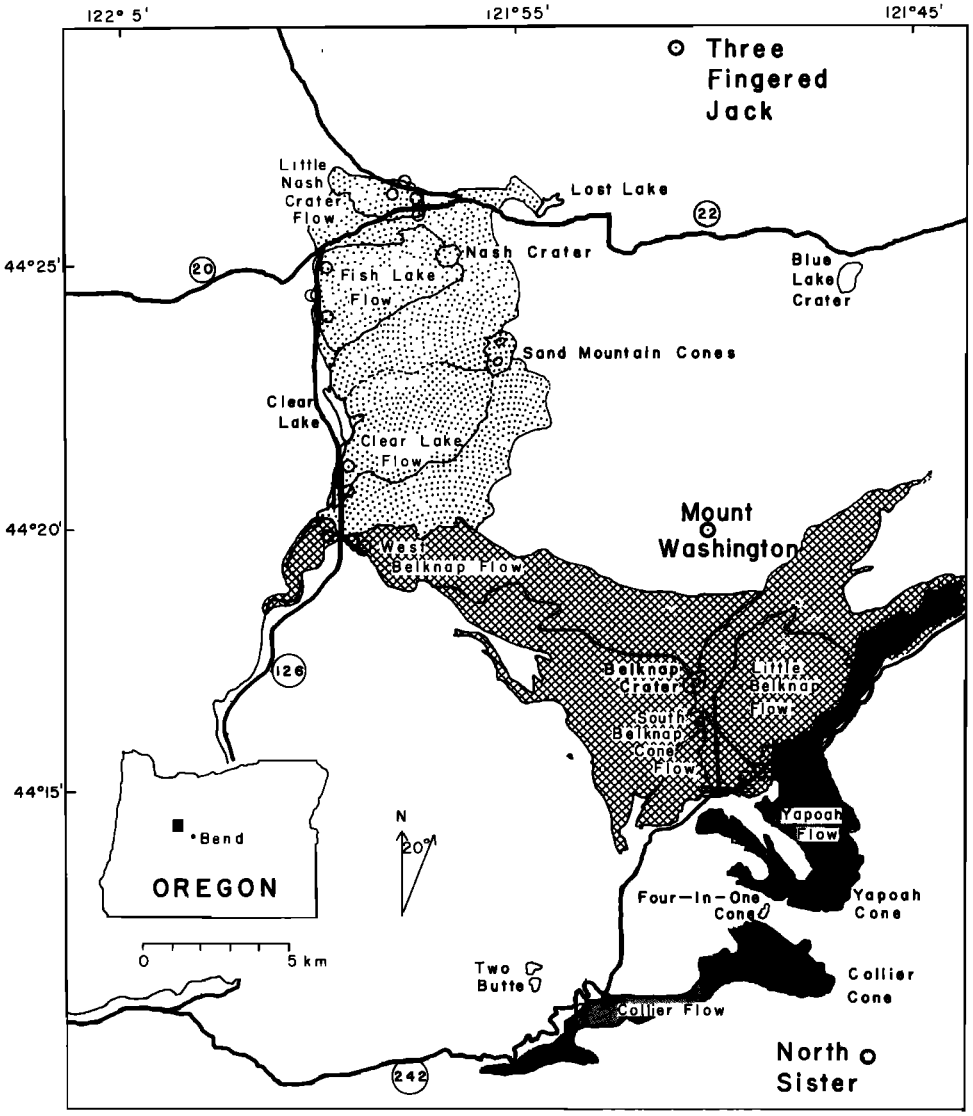


Fig. 1. Sampling site locations with respect to central High Cascades features.

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TABLE 1. Ages of Cascade Lavas

Unit	Age, years B.P.		
	Maximum	Measured	Minimum
Little Nash crater flow	1950 ± 165		300 to 400
Fish Lake flow	3850 ± 215		300 to 400
Clear Lake flow		2990 ± 380*	
		2705 ± 200†	
		3200 ± 200†	
West Belknap flow		1400 ± 100*	
		1590 ± 160*	
South Belknap cone flow		1775 ± 400*	
South Belknap flow	10,000 to 12,000		2880 ± 175
Little Belknap flow		2880 ± 175*	
Yapoah flow	2880 ± 175		2550 ± 165
Collier flow	2550 ± 165		300 to 400

*^{14}C date from *Taylor* [1967].
†^{14}C date from *Benson* [1965].

The direction of original magnetization for each sample is shown in Figures 2–4 along with the direction of magnetization after demagnetization at 200 oe. All nine flows show excellent stability upon demagnetization in fields up to 1000 oe. Only nine specimens out of a total of 94 samples had a sizable secondary component of magnetization. Two sample directions were initially removed from the main grouping of directions but moved into their respective groups upon demagnetization, three samples decreased markedly in intensity only, and four additional samples exhibited changes in direction greater than 20°. The step demagnetization experiments indicate that no major secondary components of magnetization exist in these young lava flows, with the possible exception of a small amount of viscous remanent magnetization. The mean inclination of each flow becomes more shallow, by approximately 2°–5°, after demagnetization in fields of 50–100 oe. No additional significant direction changes occur upon demagnetization in higher fields, except for a generally slight increase in dispersion of the measurements. Flow averages were calculated after demagnetization of all samples at 200 oe, since this field value was sufficient to remove all secondary components of magnetization and still have a minimal effect on the dispersion of the measurements. Five flows show relatively small increases in dispersion after demagnetization at 200 oe, whereas four flows show slight decreases in dispersion.

The observed differences in initial scatter are due mainly to differences in the type of sampling sites for each flow. For instance, in West Belknap flow, where 10 of 13 samples were collected from a large quarry, the dispersion is small (Figure 3). At the quarry all loose crustal material was graded off down to the dense, solid part of the flow. This grading reduced considerably any scatter of directions from rotation of flow materials after cooling below the Curie temperature such as that observed by *James* [1966]. In contrast, eight of the 13 cores from Little Nash crater flow were drilled from natural outcrops such as lava gutters (Figure 2). There is a general correspondence between the numbers of cores drilled from natural outcrops and the scatter that indicates that the *aa* and blocky characteristics of the flows contribute to much of the observed scatter.

The average direction of natural remanent magnetization for each flow is plotted in Figure 5. Each flow average was calculated by giving unit weight to each separately oriented core, independent of the number of specimens cut and measured. The mean directions of magnetization and precision parameters were computed by the methods described by *Fisher* [1953] and are given in Table 2. The radii of the 95% confidence level circles after demagnetization vary from 3.6° for West Belknap flow to 16.8° for South Belknap cone flow. As was mentioned earlier, the differences are related to the nature

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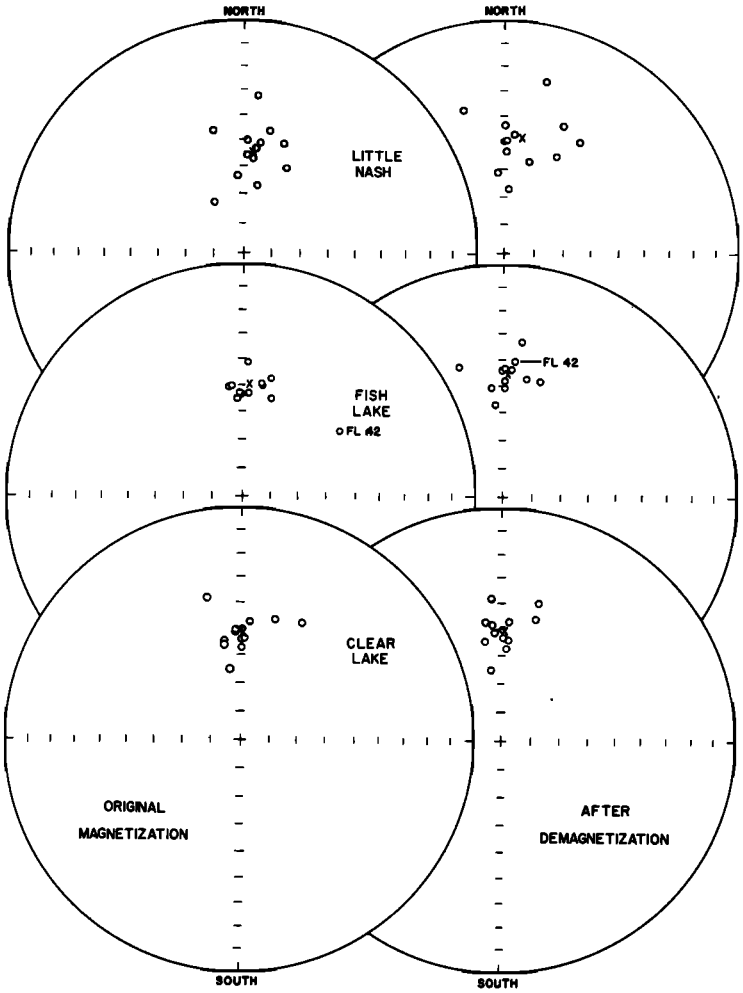


Fig. 2. Directions of remanent magnetization before and after partial demagnetization at 200 oe. North-seeking direction down, open circles; flow mean direction, cross. Lambert equal area projection.

of the sampling sites. The mean directions of magnetization are shallower than the direction of an axial dipole field. The circle of confidence for one flow encloses the axial dipole field direction. All flow means are distinct from the present geomagnetic field direction. The mean direction of magnetization of all flows, eight of which have ages between 300 and 3850 years B.P., is $D = 2.2°$, $I = +49.6°$, and $\alpha_{95} = 5.2°$. This direction is removed from the axial geocentric dipole field direction (0°, +62.9°) for the field area.

The five ^{14}C dated flows were used for a secular variation curve, since the ages of the other four flows are not known precisely enough (Figure 6). The approximate ranges and periods of the variation are as follows.

Range	Period
$D = 20°$	1100 years
$I = 23°$	1300 years

More data in the time interval would define any higher-order variation and thereby alter these values. The average values in Figure 6 were calculated by using all nine flows investigated. The declination and inclination values from 0 to 800 A.D. are similar to values obtained in the southwestern United States by

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Watanabe and DuBois [1965]. Ranges of 20° for inclination and 25° for declination given by Watanabe and DuBois are near those of 23° and 20° found in this study.

COMMENTS

Irving [1964] lists 11 paleomagnetic results covering approximately the past 7000 years. The mean declination is zero, and the inclinations are consistent with a geocentric axial dipole field. Since the inclinations of the Recent High Cascade lavas are consistently too shallow, the possibility of error in inclination due to magnetic anisotropy or geologic structure is considered.

Samples from five of the flows were heated to 800°C and cooled in a known magnetic field. The measured directions of magnetization and the direction of the known field agree within the accuracy of the experiment, approximately ±1°. No significant changes in direction are observed upon demagnetization to 1000 oe. In general, the intensity of magnetization of the remagnetized samples increased but exhibited behavior similar to that of the original magnetization upon demagnetization. Preliminary examinations of the ferromagnetic minerals indicate that the dominant opaque mineral is a titaniferous magnetite, although varying amounts of hematite are present. Individual

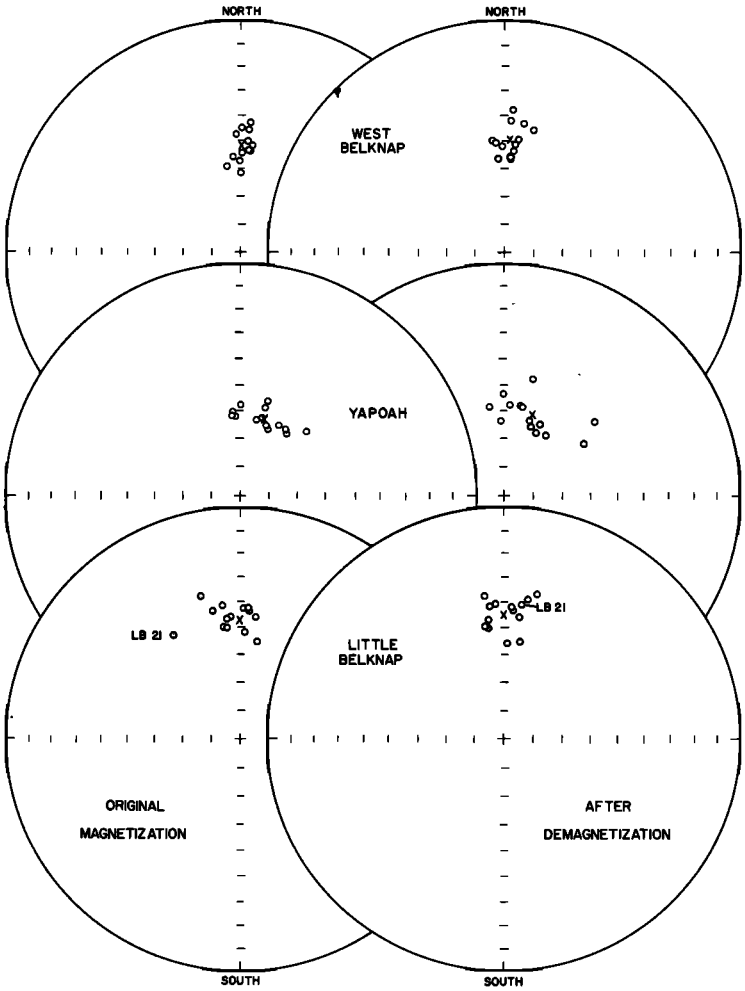


Fig. 3. Directions of remanent magnetization before and after partial demagnetization at 200 oe. Symbols used are those in Figure 2.

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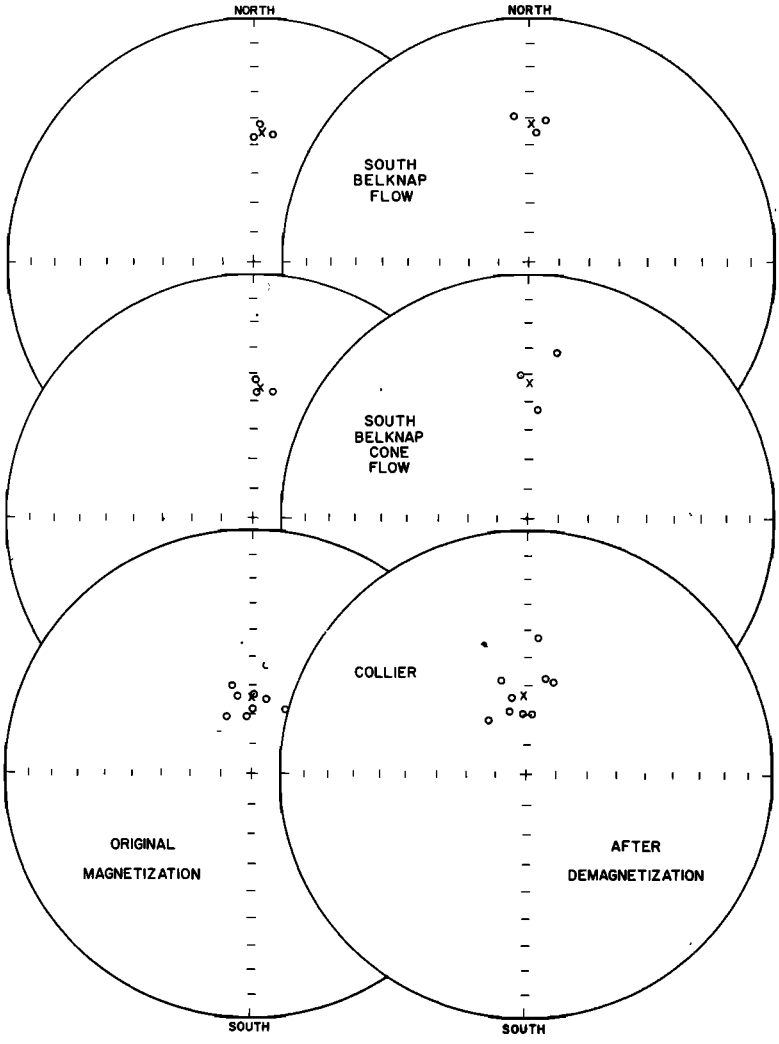


Fig. 4. Directions of remanent magnetization before and after partial demagnetization at 200 oe. Symbols used are those in Figure 2.

polished sections include a range of titanomagnetite grains from homogeneous unaltered types to those with numerous exsolution lamellas of ilmenite. One might question the directions of magnetization of specimens with only low-temperature oxidation states, but the other specimens should give stable directions [*Larson et al.*, 1969].

The maximum vertical relief between the source and the terminal lobes of several flows is approximately 700 meters. The orientations of the samples were not adjusted for the local dip of the lava flows, since the flow attitudes appear to be initial dips [*Taylor*, 1967]. The within-flow agreement of magnetization directions from sites with different apparent dips, 0°–10°, also indicates initial dips. Finally, hypothetical dip corrections would have a predominant east-west sense and would not remove the shallow magnetic inclinations. Since neither structure nor magnetic parameters suggests anomalous directions of magnetization, the shallow inclinations probably represent geomagnetic field directions.

SUMMARY AND CONCLUSIONS

A detailed comparison of this study with most other individual investigations will not be at-

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tempted. The geographical separations, limited data, and imperfect knowledge of the behavior of the Recent geomagnetic field would make such comparisons tenuous at this time. The results do suggest that the geomagnetic field fluctuated about a mean direction removed from an axial dipole direction (Figure 7). *Doell* [1969] and *Doell and Cox* [1971] present results from a study of 54 lavas from the Kau volcanic series on Hawaii. The age range of the series, using reasonable geologic and paleomagnetic constraints, is estimated to be a 'period of several thousands of years, occurring sometime after 10,000 years ago' [*Doell*, 1969, p. 4857]. The minimum age for the youngest flow is estimated to be no less than several hundred years ago. The mean virtual geomagnetic pole position for the Kau series is 81.5°N, 61.3°E with a cone of confidence (α_{95}) of 0.78° and is considered to be a main dipole field direction [*Doell*, 1969]. The Recent High Cascade lavas give a virtual geomagnetic pole of 76.1°N, 50.3°E with an α_{95} of 5.2°. The agreement of the mean pole positions supports Doell's conclusion that this mean direction is a significant departure of the main geomagnetic field from an axial direction.

Wilson [1970, 1971] proposes a persistent northward displacement of the effective dipole source of the main field over Quaternary and Recent times (about the past 2 m.y.). The effect of this northward displacement of the main dipole on paleomagnetic pole positions calculated by assuming a centered geocentric axial dipole source is to systematically displace poles along a great circle connecting the field area and the geographic pole to the far side of the geographic poles. The mean displacement of 28 Quaternary and Recent pole determinations is 2.9° away from the geographic pole [*Wilson*, 1971]. Thus, although the results of this study are in better absolute agreement with Doell's results (13.9° and 8.5° displacements, respectively), the sense of the offset supports Wilson's proposal for a displaced main geomagnetic dipole source.

Geomagnetic field variations range from periods of less than 10 years to periods greater than 10^6 years. Periods associated with secular variation due to westward drift of the nondipole field and the growth and decay of individual features are in the range 10^2–10^3 years. Changes in the orientation and intensity of the main dipole contribute to the spectrum at about 10^4 years [*Cox and Doell*, 1964]. Thus the Recent High Cascade lavas may represent a period of time long enough to have averaged secular

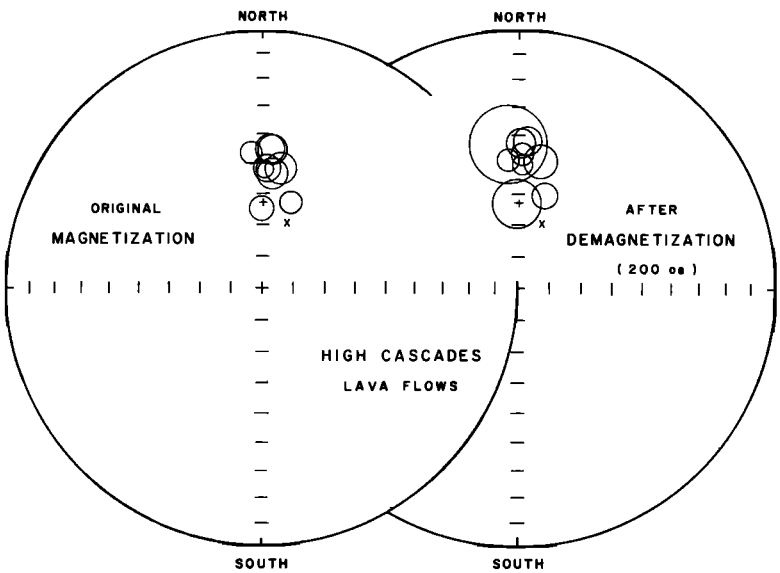


Fig. 5. Flow average directions of magnetization and cones of confidence for original magnetization and directions after demagnetization. Axial dipole field direction, plus sign; present geomagnetic field direction, cross.

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TABLE 2. Flow Average Directions of Magnetization

Flow	Experiment*	I, deg	D, deg	J(avg), emu/cm^3	Exponent	N	R	κ	α_{95}	ASD	θ	ϕ
				Original Magnetization								
Clear Lake	NRM	51.9	2.0	1.05 ± 1.43	-2	13	12.798	59.5	5.4	10.5	78.1	49.8
Collier	NRM	64.8	359.7	1.06 ± 0.61	-2	9	8.915	94.2	5.3	8.3	87.4	233.8
Fish Lake	NRM	51.4	8.3	8.07 ± 5.82	-3	12	11.779	49.8	6.2	11.5	76.1	27.4
Little Belknap	NRM	46.3	354.8	8.85 ± 8.65	-3	14	13.819	71.9	4.7	9.6	72.8	73.9
Little Nash	NRM	53.5	5.1	1.29 ± 0.79	-2	13	12.719	42.6	6.4	12.4	78.9	35.7
South Belknap cone	NRM	45.1	3.3	5.48 ± 1.68	-3	3	2.995	417.0	6.0	4.0	72.2	48.4
South Belknap flow	NRM	45.2	3.1	1.88 ± 0.36	-2	3	2.995	408.7	6.1	4.0	72.3	48.9
West Belknap	NRM	52.1	0.7	6.53 ± 2.48	-3	13	12.927	164.7	3.2	6.3	78.3	55.0
Yapoah	NRM	61.7	17.9	1.87 ± 1.04	-2	14	13.822	73.1	4.7	9.5	77.0	328.1
				Demagnetized								
Clear Lake	200 oe	48.8	355.5	5.69 ± 9.68	-3	13	12.851	80.7	4.6	9.0	75.0	73.3
Collier	200 oe	63.0	359.1	6.09 ± 3.69	-3	8	7.768	30.2	10.2	14.8	89.3	173.0
Fish Lake	200 oe	46.9	1.3	6.05 ± 8.79	-3	12	11.892	102.1	4.3	8.0	73.7	54.0
Little Belknap	200 oe	42.5	0.6	7.10 ± 8.62	-3	8	7.921	88.2	5.9	8.6	70.4	56.6
Little Nash	200 oe	48.5	9.2	5.41 ± 5.07	-3	13	12.654	34.7	7.1	13.7	73.3	29.0
South Belknap cone	200 oe	43.6	356.4	2.12 ± 0.64	-3	3	2.964	55.0	16.8	10.9	71.0	68.1
South Belknap flow	200 oe	42.6	3.1	1.13 ± 0.46	-2	3	2.996	462.2	5.7	3.8	70.3	49.7
West Belknap	200 oe	50.0	1.8	4.01 ± 2.55	-3	13	12.909	131.7	3.6	7.1	76.4	51.5
Yapoah	200 oe	59.2	15.3	8.15 ± 6.10	-3	14	13.783	60.0	5.2	10.5	77.9	343.3
				Section averages								
All flows	NRM	52.6	3.5	1.12 ± 0.48	-2	9	8.925	106.6	5.0	7.8	78.6	43.2
All flows	200 oe	49.6	2.2	6.21 ± 2.57	-3	9	8.919	98.6	5.2	8.2	76.1	50.3

*Intensity of demagnetizing field or original magnetization.

I, inclination of flow mean direction; D, declination of flow mean direction; J(avg), intensity of magnetization; N, number of separately oriented samples; R, vector sum of N unit vectors; κ, intraflow precision parameter [*Fisher*, 1953]; α_{95}, radius of 95% confidence interval about direction of R; ASD, angular standard deviation of specimen directions; θ, latitude of virtual geomagnetic pole; ϕ, longitude of virtual geomagnetic pole.

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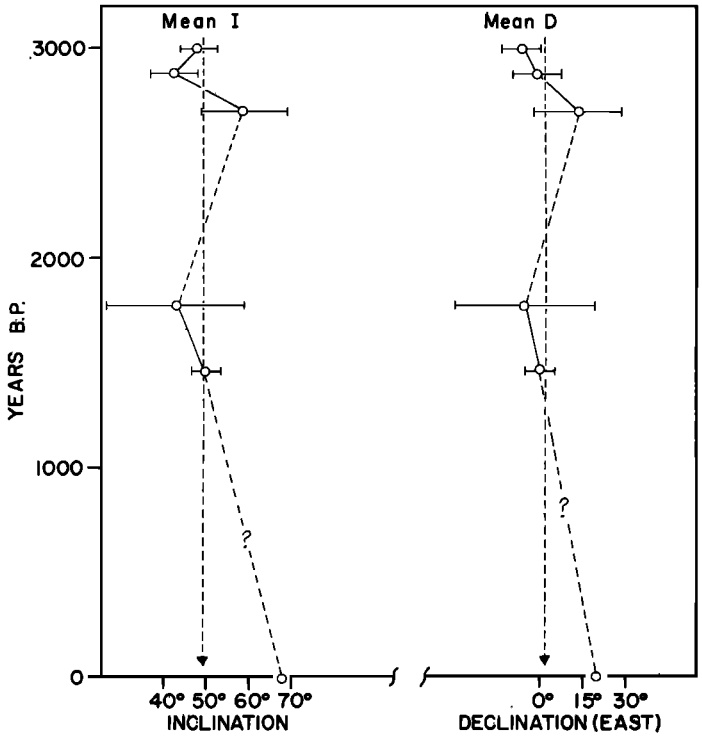


Fig. 6. Secular variation curves of inclination and declination versus years before present for five dated flows with 95% confidence limits.

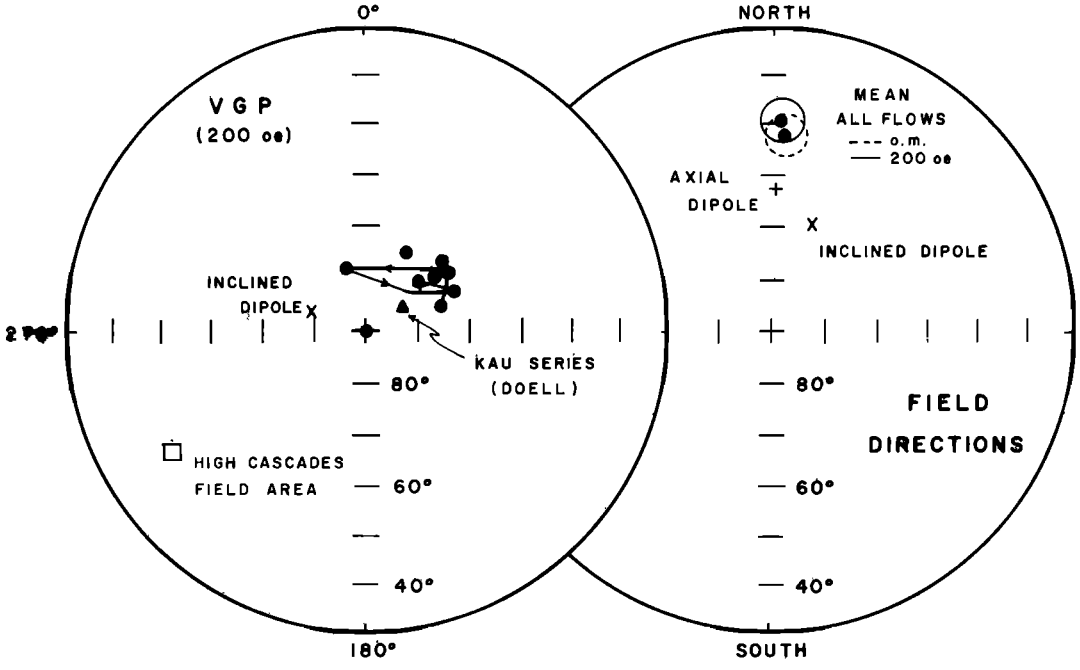


Fig. 7. Section average directions of magnetization before and after demagnetization and associated virtual geomagnetic pole (VGP) positions for the High Cascade lavas. Solid triangle represents mean VGP position from Kau series [*Doell*, 1969]. Line with arrows connects VGP positions for flows in Figure 6. Parts of Lambert equal area projections.

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changes due to the nondipole field but not long enough to include a change in the orientation of the main dipole. This conclusion must be considered tentative, as additional data are needed from other rocks of the same age before the significance of the directions can be determined.

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